

Experiment 1

Exp. Name : Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

Name : Vineet Kachare

PRN : B25ET1091

Batch: SY2

```
#include <iostream>
using namespace std;
void inputArray(int a[], int n) {
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
}
void displayArray(int a[], int n) {
    cout << "Array elements are: ";
    for (int i = 0; i < n; i++) {
        cout << a[i] << " ";
    }
    cout << endl;
}
void swapElements(int &x, int &y) {
    int temp = x;
    x = y;
    y = temp;
}
Void sortArray(int a[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                swapElements(a[j], a[j + 1]);
            }
        }
    }
}
int main() {
    int n;
    cout << "Enter the size of array: ";
    cin >> n;
    int a[n];
    inputArray(a, n);
    cout << "\nBefore Sorting:\n";
    displayArray(a, n);
    sortArray(a, n);
}
```

```
cout << "\nAfter Sorting:\n";  
displayArray(a, n);  
return 0;
```